

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 23	
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Batch	2025-29
Class	CSE A2
Academic Year & Semester	2026-27, III
Date of Submission	27/10/26
Title of Assignment:	Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.

Practice questions
 	
Output Screenshots Solve questions on given link and attach screen shot of leetcode with your login and o/p
	1. Create a program to display database records one by one using ResultSet
	2. Create a Java program that retrieves employee records from a database and displays them one by one using ResultSet. Show employee ID, name, department, and salary sequentially
	import java.sql.*;
	import java.util.Scanner;
	
	public class EmployeeCRUD {
	
	public static void main(String[] args) {
	
	String url = "jdbc:mysql://localhost:3306/companydb";

```
String username = "root";
String password = "arpit@2006";

try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection con =
        DriverManager.getConnection(url, username, password);

    Scanner sc = new Scanner(System.in);

    while (true) {

        System.out.println("\n===== Employee Record Management =====");
        System.out.println("1. Insert");
        System.out.println("2. Display");
        System.out.println("3. Update");
        System.out.println("4. Delete");
        System.out.println("5. Exit");
        System.out.print("Enter choice: ");

        int choice = sc.nextInt();

        if (choice == 1) {

            System.out.print("Enter Employee ID: ");
            int id = sc.nextInt();
            sc.nextLine();

            System.out.print("Enter Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Department: ");
            String department = sc.nextLine();
```

```
            System.out.print("Enter Salary: ");
            double salary = sc.nextDouble();

            PreparedStatement ps = con.prepareStatement(
                "INSERT INTO employee VALUES (?, ?, ?, ?)"
            );

            ps.setInt(1, id);
            ps.setString(2, name);
            ps.setString(3, department);
            ps.setDouble(4, salary);

            ps.executeUpdate();

            System.out.println("Employee inserted successfully.");
        }

        else if (choice == 2) {

            Statement stmt = con.createStatement();

            ResultSet rs =
                stmt.executeQuery("SELECT * FROM employee");

            System.out.println("\nID\tName\tDepartment\tSalary");

            while (rs.next()) {

                System.out.println(
                    rs.getInt("employee_id") + "\t" +
                    rs.getString("name") + "\t" +
                    rs.getString("department") + "\t\t" +
                    rs.getDouble("salary")
```

```
            );
        }
    }

    else if (choice == 3) {

        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();

        System.out.print("Enter New Salary: ");
        double salary = sc.nextDouble();

        PreparedStatement ps = con.prepareStatement(
            "UPDATE employee SET salary = ? WHERE employee_id = ?"
        );

        ps.setDouble(1, salary);
        ps.setInt(2, id);

        ps.executeUpdate();

        System.out.println("Employee updated successfully.");
    }

    else if (choice == 4) {

        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();

        PreparedStatement ps = con.prepareStatement(
            "DELETE FROM employee WHERE employee_id = ?"
        );

        ps.setInt(1, id);
```

	ps.executeUpdate();
	
	System.out.println("Employee deleted successfully.");
	}
	
	else if (choice == 5) {
	
	System.out.println("Exiting...");
	break;
	}
	
	else {
	System.out.println("Invalid choice.");
	}
	}
	
	con.close();
	sc.close();
	
	} catch (Exception e) {
	System.out.println("Error: " + e.getMessage());
	}
	}
	}

Output Screenshots (if applicable)	

```
PS C:\Users\Arpit\OneDrive\Desktop\JDBC_project>
PS C:\Users\Arpit\OneDrive\Desktop\JDBC_project>  c:; cd 'c:\Users\Arpit\OneDrive\Desktop\JDBC_project'; & 'c:\Users\
Arpit\.vscode\extensions\redhat.java-1.58.0-win32-x64\jre\25.0.4.1-win32-x86_64\bin\java.exe' '@C:\Users\Arpit\AppDat
a\Local\Temp\cp_4a4cvajz19gmhlalggj62y0q0.argfile' 'EmployeeCRUD'

===== Employee Record Management =====
1. Insert
2. Display
3. Update
4. Delete
5. Exit
Enter choice: 2

ID      Name    Department      Salary
101     Arpit   CSE             100000.0

===== Employee Record Management =====
1. Insert
2. Display
3. Update
4. Delete
5. Exit
Enter choice: 1
Enter Employee ID: 102
Enter Name: Aryan
Enter Department: AIML
Enter Salary: 50000
Employee inserted successfully.

===== Employee Record Management =====
1. Insert
2. Display
3. Update
4. Delete
5. Exit
Enter choice: 3
Enter Employee ID: 101
Enter New Salary: 150000
Employee updated successfully.
```

```
===== Employee Record Management =====
1. Insert
2. Display
3. Update
4. Delete
5. Exit
Enter choice: 2

ID      Name    Department      Salary
101     Arpit   CSE             150000.0
102     Aryan   AIML            50000.0

===== Employee Record Management =====
1. Insert
2. Display
3. Update
4. Delete
5. Exit
Enter choice:
```

Conclusion	Thus, the Java application was successfully developed using JDBC to manage employee records in a MySQL database. The program performs insert, display, update, and delete operations. The ResultSet interface is used to retrieve and navigate employee records one by one, displaying the employee ID, name, department, and salary. This practical helped in understanding database connectivity and record management using JDBC.